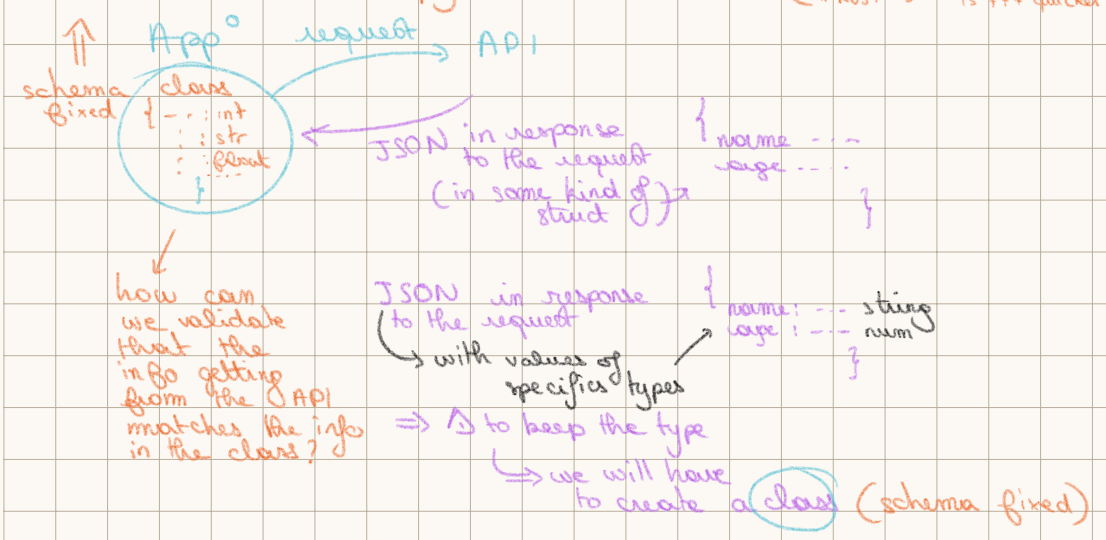
PYDANTIC (Data Validation) => python

→ the most widely used data validation library for python
→ fast & extensible.

FAST API ← pydantic model used +++
data validation => pydantic ⟶ RUST (back end written in RUST) => process of data valid° is +++ quicker



Langgraph = framework where we ++ complex workflows

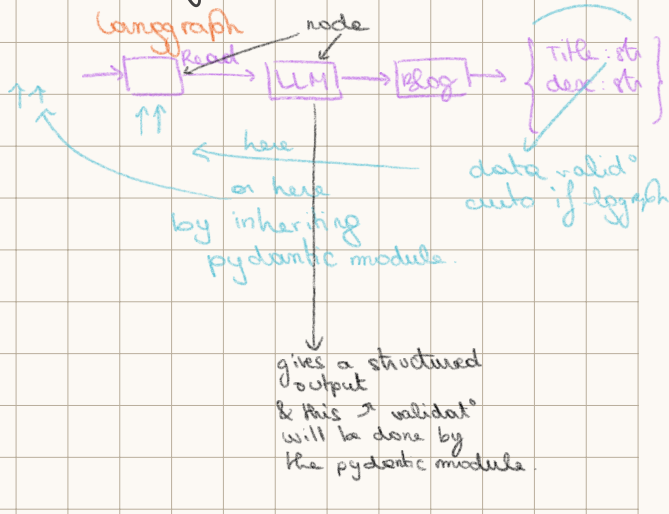
ex of workflow for a cert^N task: upload youtube video :

youtube —convert into→ Blog

=> as soon as I upload yt video
→ some workflow = triggered
where I read contxt of yt vid.
→ next wf: LLM called with that info
=> LLM ++ blog

process can be auto. with langgraph

Whenever node executed



LANGGRAPH + LANGSMITH

track => metrics => langgraph studio

① AI agents vs Agentic AI

② langgraph (langchain)
③ Build simple workflow using langgraph (code) → graph, node, functions

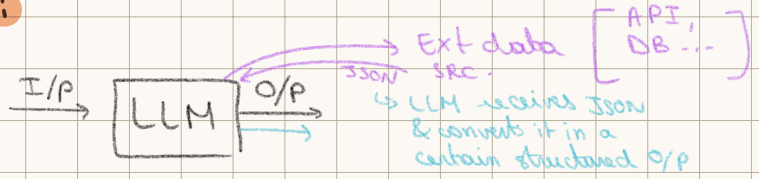
AI agents

→ refers to some software prog designed to perform specific task without human intervention with a degree of autonomy

Agentic AI System

→ frameworks where multiple AI agents can collaborate and make decis° independently to achieve a larger goal.

→ **Ex of AI agents :**



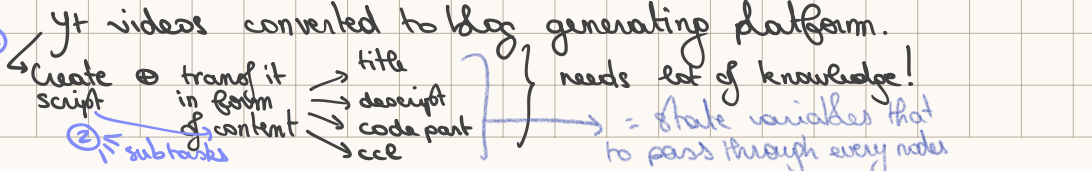
1 simple II task : when I give I/P LLM gives O/P

if I give an input will I receive an output?
↳ I/P = "hello ..." → yes
↳ I/P = "provide me the recent news for today" → no bc LLM don't have that specific data

=> we try to train LLM with external source
→ LLM is working IIthy & redirecting a query to an external data src & getting an o/p
↳ a degree of autonomy.

→ **Ex of Agentic AI system:** → Broader framework
→ instead of 1 AI agent which does a specific task here multiple AI agents able to communicate with each other to solve a bs statment each a specific goal.

use case : upload lot of yt videos
→ If I want to dev app :
yt videos converted to blog generating platform.

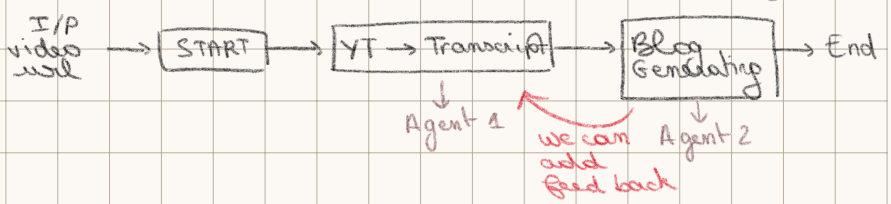


=> we can convert this entire work & give it to LLM

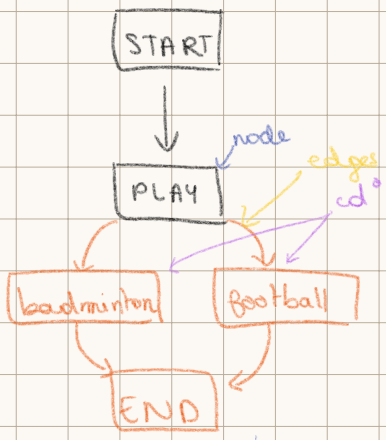
① YT video → transcript [code] => Langgraph → YT API } AI 1st Agent

② transcript convert → title, desc, code, ccc } AI 2nd Agent ← 1st passes its O/P to 2nd

=> ++ workflow to solve a goal



Coding: (workflow)



I want to automate using Langgraph.

Components:

① □ = node
↳ = python f°
↳ = task

② → = edges connecting node
= cond° edge

③ State (like ...) = state schema = I/P for all nodes

④ State Graph = structure of the entire graph

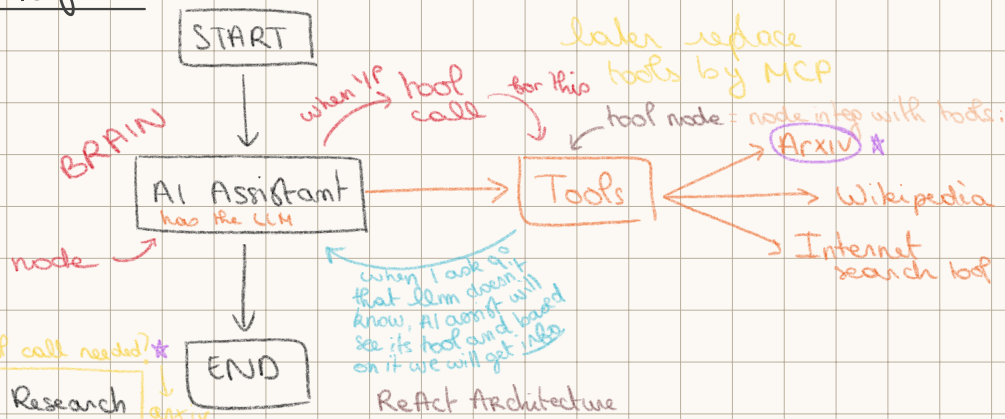
CHATBOT WITH LANGGRAPH

① Chatbot with Langgraph + Tools {Multiple tools}

② ReAct Architecture [Reasoning + Act]

MCP
⇓
(Model Context Protocol)

Chatbot + Workflow



① 1st part → tool call needed
tell me about Research paper [name]
then search for AI agent
→ ② 2nd part can LLM provide the answer? (no need tool)

③ combining ① + ② answers

Can the Chatbot + LLM will answer to:

" provide me the recent AI news "

↳ NO!

↳ LLM ~~(outdated info)~~

